Output Ejercicio 8

Foto terminal con el hexadecimal:

```
TERMINAL
0x0B,0x01,0x00,0x07,0x09,0x00,0x00,0x01,0x04,0x0B,0x0B,0x04,0x01,0x01,0x01,0x03,
0x03,0x06,0x01,0x03,0x19,0x3F,0x3F,0x42,0x45,0x42,0x42,0x2F,0x0B,0x04,0x1C,0x29,
0x1C,0x19,0x13,0x06,0x07,0x10,0x23,0x0D,0x00,0x00,0x02,0x13,0x1C,0x1C,0x20,0x20,
0x20,0x23,0x29,0x23,0x16,0x0B,0x07,0x10,0x29,0x26,0x20,0x19,0x13,0x07,0x04,0x04,
0x57,0x66,0x62,0x48,0x62,0x53,0x69,0x7C,0x8D,0x29,0x20,0x16,0x19,0x1C,0x1C,0x1C,
0x0D,0x10,0x29,0x92,0x8F,0x8A,0x90,0x7F,0x7A,0x36,0x13,0x06,0x07,0x03,0x06,0x0B,
0x02,0x01,0x07,0x07,0x00,0x00,0x02,0x07,0x0B,0x09,0x03,0x00,0x00,0x01,0x02,0x03,
0x03,0x01,0x00,0x02,0x06,0x3C,0x48,0x48,0x48,0x42,0x42,0x29,0x09,0x09,0x23,0x20,
0x16,0x20,0x0D,0x07,0x10,0x1C,0x23,0x03,0x00,0x00,0x04,0x13,0x20,0x20,0x1C,0x20,
0x1C,0x1C,0x26,0x26,0x19,0x0D,0x09,0x0D,0x23,0x23,0x1C,0x19,0x10,0x07,0x06,0x03,
END_EQUALIZED_HEX
I (17730) main_task: Returned from app_main()

Done

====== IMAGEN ORIGINAL ======
0x87,0x88,0x95,0x9E,0x9F,0x9F,0xA1,0xA4,0xA5,0xA5,0xA5,0xA6,0xA5,0xA6,0xA8,0xAB,0xAB,0xAB,0xAA,0xAA,0xAC,0xA7,0x94,0x85,0xA0,0xC9,0xE9,0xE4,0xDC,0xC9,0xAD,0x
9A,0x9D,0x9E,0x96,0x82,0x60,0x54,0x53,0x52,0x51,0x58,0x45,0x3A,0x3A,0x3A,0x3A,0x3B,0x3A,0x39,0x39,0x39,0x39,0x39,0x3C,0x3E,0x42,0x69,0x9D,0x99,0x75,0x56,0x4B
,0x4E,0x60,0x65,0x85,0xB7,0xA3,0x71,0x67,0x63,0x96,0xBB,0xB9,0xA6,0x8A,0x76,0x75,0x81,0x99,0xAF,0xAE,0xA9,0xA1,0x97,0x83,0x4B,0x45,0x50,0x51,0x4E,0x42,0x3C,0
x3D,0x3D,0x9D,0x99,0x8C,0x87,0x92,0x9E,0xA1,0xA2,0xA5,0xA8,0xA9,0xA9,0xA9,0xAA,0xA9,0xA9,0xAB,0xAD,0xB0,0xB0,0xB6,0xBB,0xCE,0xB9,0x9B,0x88,0x93,0xBD,0xDD,0xD
E,0xDC,0xD4,0xB7,0x9E,0x94,0x74,0x5B,0x5C,0x57,0x50,0x56,0x50,0x3A,0x39,0x38,0x38,0x39,0x3A,0x39,0x39,0x38,0x37,0x35,0x38,0x4D,0x41,0x3E,0x3F,0x42,0x6D,0x9F,
0x94,0x7B,0x61,0x67,0x66,0x64,0x64,0x8E,0xBC,0xB1,0x76,0x94,0xBF,0xC1,0xC1,0xBC,0xA5,0x89,0x76,0x79,0x86,0x9E,0xAD,0xA8,0x92,0x50,0x48,0x51,0x51,0x4D,0x41,0x
3D,0x3D,0x3D,0x3C,0xA0,0x9E,0x9E,0x9F,0x95,0x8A,0x8F,0x9F,0xA5,0xA7,0xA9,0xAC,0xAD,0xAC,0xAB,0xAC,0xAC,0xAD,0xB3,0xBA,0xC6,0xCE,0xCF,0xC6,0xBC,0xB5,0xA2,0x8C
,0x8C,0xB5,0xDF,0xE3,0xDF,0xD8,0xA3,0x5F,0x5F,0x5F,0x4D,0x50,0x58,0x3F,0x39,0x38,0x37,0x37,0x37,0x39,0x39,0x38,0x37,0x35,0x35,0x3D,0x55,0x61,0x3E,0x3D,0x3D,0
x3E,0x43,0x70,0xA7,0xA2,0x77,0x6B,0x66,0x5F,0x5B,0x61,0x93,0xB8,0xAA,0xC2,0xC7,0xC7,0xC6,0xC4,0xBC,0xA2,0x88,0x78,0x7D,0x89,0x8F,0x4F,0x4B,0x53,0x51,0x4D,0x4
0,0x3D,0x3D,0x3D,0x3D,0x40,0xA3,0xA2,0xA1,0xA2,0xA2,0xA2,0x9D,0x91,0x8D,0x9C,0xA6,0xA9,0xAC,0xAE,0xB0,0xAF,0xAF,0xB0,0xB4,0xBB,0xC8,0xD1,0xCD,0xC4,0xBD,0xB9,
0xB8,0xB7,0xA8,0x93,0x8E,0xB9,0xDE,0xE0,0x7A,0x5C,0x61,0x50,0x49,0x56,0x48,0x3C,0x37,0x3D,0x3B,0x37,0x36,0x36,0x36,0x37,0x35,0x33,0x3A,0x3D,0x3D,0x42,0x3C,0x
3B,0x3A,0x3B,0x3C,0x3E,0x45,0x79,0xAE,0x9A,0x72,0x60,0x5A,0x57,0x56,0x61,0x9B,0xC2,0xCE,0xCD,0xCB,0xC9,0xC7,0xC3,0xB6,0x9E,0x83,0x6F,0x4A,0x4C,0x54,0x53,0x4C
,0x40,0x3E,0x3E,0x3E,0x3D,0x40,0x46,0xA4,0xA5,0xA5,0xA5,0xA4,0xA5,0xA5,0xA7,0xA5,0x99,0x90,0x97,0xA6,0xAA,0xAC,0xB0,0xB2,0xB5,0xB7,0xBE,0xC8,0xCE,0xC8,0xC0,0
xBD,0xBA,0xBA,0xBA,0xBB,0xBC,0xB2,0x9B,0x8B,0x90,0x59,0x5C,0x5A,0x46,0x4A,0x51,0x3E,0x3C,0x3A,0x40,0x46,0x42,0x39,0x35,0x35,0x34,0x32,0x36,0x3D,0x3C,0x3C,0x3
B,0x3A,0x3A,0x3A,0x3A,0x3A,0x3B,0x3C,0x40,0x4B,0x83,0xAD,0x8E,0x63,0x5A,0x58,0x57,0x73,0xA0,0xB4,0xCC,0xD1,0xCE,0xCA,0xC7,0xC5,0xC0,0x95,0x4B,0x4E,0x55,0x53,
Juan*  0 0  Compilación  Update available, click to download.  CristobalDB (1 week ago)  Tab Moves Focus  Ln 39, Col 26  Spaces: 4  UTF-8  LF  { } C  ESP-I
```

Foto original tomada con ESP-CAM:

Foto después de la ecualización:

take_picture.c

Programa principal que corre en el ESP32. Contiene:

- *camera_config* — estructura con la configuración del hardware: pines, resolución 96×96, escala de grises, buffer en DRAM
- *app_main()* — función de entrada, ejecuta todo en orden:
- Inicializa la cámara
- Espera 500ms para estabilizar el sensor
- Captura un frame con esp_camera_fb_get()
- Espera 5000ms para que el monitor serial esté conectado
- Imprime la imagen original en hex (16 bytes por línea, 10ms entre líneas para no saturar el UART), entre tags BEGIN_IMAGE_HEX / END_IMAGE_HEX
- Llama a equalize_histogram(), imprime el resultado entre tags BEGIN_EQUALIZED_HEX / END_EQUALIZED_HEX
- Libera el buffer de la cámara

equalize.c
Módulo de procesamiento de imagen. Contiene:

- *equalize_histogram(input, output, len)* — toma la imagen cruda y produce una ecualizada:
- Calcula el histograma: cuenta cuántos píxeles tienen cada valor (0–255)
- Calcula la CDF: suma acumulada del histograma
- Encuentra cdf_min: primer valor no cero de la CDF
- Construye una tabla de transformación que redistribuye los valores para maximizar el contraste
- Aplica la tabla a cada píxel del buffer de salida

equalize.h
Interfaz pública de equalize.c. Declara la firma de equalize_histogram() para que take_picture.c pueda usarla sin conocer su implementación.

CMakeLists.txt
Archivo de build del proyecto. Le dice al sistema de compilación:

- Qué archivos compilar: take_picture.c y equalize.c
- Dónde buscar headers: directorio actual (.)
- Qué librería externa necesita: esp32-camera

flash_and_dump.sh
Script que orquesta todo desde la PC:

- Activa el entorno ESP-IDF
- Limpia /tmp/serial_output.txt para evitar datos viejos
- Flashea el ESP32 y abre el monitor con script, que muestra el output en pantalla y lo guarda en el archivo simultáneamente
- Al salir con Ctrl+T X, extrae automáticamente con sed los bloques hex de cada imagen y los imprime listos para copiar al Colab.

Para tomar la foto hay que seguir los siguientes pasos:

1. *cd ~/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_1/Ejercicio_8 && ./flash_and_dump.sh* : con este comando flasheamos el ESP32 y abrimos el monitor serial para ver la salida.
2. *Ctrl T+X* : Con este comando cerramos el monitor y se muestran los datos x0 completos de las imágenes, en la salida de la terminal.
3. Por último copiamos la salida para la imagen original y la imagen ecualizada, donde ambas están en la salida de la terminal, correctamente diferenciadas.